Practical-8: Intermediate Programs Using Pointer

1) Swap value of two numbers using pointer. (A)

Program :-

```
#include <stdio.h>
void main(){
	int n1,n2;
	printf("Enter the First Number : ");
	scanf("%d",&n1);
	printf("Enter the Second Number : ");
	scanf("%d",&n2);
	int *ptr1 = &n1;
	int *ptr2 = &n2;
	int temp = *ptr1;
	*ptr1 = *ptr2;
	*ptr2 = temp;
	printf("n1 = %d\n",n1);
	printf("n2 = %d\n",n2);
}
```

Output :-

```
Enter the First Number : 12
Enter the Second Number : 34
n1 = 34
n2 = 12
```

2) Store n elements in an array and print the elements using pointer.(A)

Program :-

```
	#include <stdio.h>
	void main(){
		int n = 0,i = 0;
		printf("Enter the Size of Array : ");
		scanf("%d",&n);
		int arr[n];
		for(i=0;i<n;i++){
			printf("Enter the value at Index %d :",i);
			scanf("%d",&arr[i]);
		}
		//Print Array
		printf("\nArray : \n");
		int *ptr = arr;
		for(i=0;i<n;i++){
			printf("%d ",*(ptr+i));
		}
```

Output :-

```
Enter the Size of Array : 5
Enter the value at Index 0 :23
Enter the value at Index 1 :45
Enter the value at Index 2 :67
Enter the value at Index 3 :89
Enter the value at Index 4 :23

Array :
23 45 67 89 23
```

3) Find even and odd numbers in array using pointer. (A)

Program :-

```
#include <stdio.h>
void main(){
        int n = 0,i = 0;
        printf("Enter the Size of Array : ");
        scanf("%d",&n);
        int arr[n];
        for(i=0;i<n;i++){
                printf("Enter the value at Index %d :",i);
                scanf("%d",&arr[i]);
        }
        int *ptr = arr;
        //Print Even Numbers
        printf("\nEven Numbers : \n");
        for(i=0;i<n;i++){
                if( *(ptr+i)%2 == 0){
                        printf("%d ",*(ptr+i));
                }
        }
        //Print Odd Numbers
        printf("\nOdd Numbers : \n");
        for(i=0;i<n;i++){
                if( *(ptr+i)%2 != 0){
                        printf("%d ",*(ptr+i));
                }
        }
}
```

Output :-

```
Enter the Size of Array : 5
Enter the value at Index 0 :1
Enter the value at Index 1 :2
Enter the value at Index 2 :3
Enter the value at Index 3 :4
Enter the value at Index 4 :5

Even Numbers :
2 4
Odd Numbers :
1 3 5
```

4) Print positive and negative numbers in array using pointers. (A)

Program :-

```
#include <stdio.h>
void main(){
        int n = 0,i = 0;
        printf("Enter the Size of Array : ");
        scanf("%d",&n);
        int arr[n];
        for(i=0;i<n;i++){
                printf("Enter the value at Index %d :",i);
                scanf("%d",&arr[i]);
        }
        int *ptr = arr;
        //Print Positive Numbers
        printf("\nPositive Numbers : \n");
        for(i=0;i<n;i++){
                if( *(ptr+i) >= 0){
                        printf("%d ",*(ptr+i));
                }
        }
        //Print Negative Numbers
        printf("\nNegative Numbers : \n");
        for(i=0;i<n;i++){
                if( *(ptr+i) < 0){
                        printf("%d ",*(ptr+i));
                }
        }
}
```

Output :-

```
Enter the Size of Array : 5
Enter the value at Index 0 :12
Enter the value at Index 1 :-23
Enter the value at Index 2 :-12
Enter the value at Index 3 :34
Enter the value at Index 4 :-9

Positive Numbers :
12 34
Negative Numbers :
-23 -12 -9
```

5) Sort array using pointers.(B)

Program :-

```
#include <stdio.h>
void main(){
	int n = 0,i = 0,j = 0;
	printf("Enter the Size of Array : ");
	scanf("%d",&n);
	int arr[n];
	for(i=0;i<n;i++){
		printf("Enter the value at Index %d :",i);
		scanf("%d",&arr[i]);
	}
	//Sort Array By Selection Sort
	int *ptr = arr;
	int minIdx = 0,temp = 0;
	for(i=0;i<n-1;i++){
		minIdx = i;
		for(j=i+1;j<n;j++){
			if(*(ptr+j) < *(ptr+minIdx)){
				minIdx = j;
			}
		}
		//swap the number
		if(minIdx != i){
			temp = *(ptr+i);
			*(ptr+i) = *(ptr+minIdx);
			*(ptr+minIdx) = temp;
		}
	}
```

```
        //Print Array
        printf("\nArray : \n");
        for(i=0;i<n;i++){
                printf("%d ",*(ptr+i));
        }
}
```

Output :-

```
Enter the Size of Array : 10
Enter the value at Index 0 :12
Enter the value at Index 1 :34
Enter the value at Index 2 :56
Enter the value at Index 3 :12
Enter the value at Index 4 :14
Enter the value at Index 5 :1
Enter the value at Index 6 :78
Enter the value at Index 7 :65
Enter the value at Index 8 :32
Enter the value at Index 9 :11

Array :
1 11 12 12 14 32 34 56 65 78
```